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PROBLEM

Engineers transcribe values from datasheets, P&ID drawings, and control-system manuals into spreadsheets and simulators by hand. Free-text RAG hides the citation; one mis-transcribed parameter lands in a safety calculation with no way back to the source page.

ARCHITECTURE

Two MCP services over three folders on disk:

- DOCS — bronze (raw PDFs) → silver (page layout + text) → gold (regions with page + bbox).
- CANVASES — one folder per canvas. Share like a file.
- SESSION — live state in memory; the same JSON drives every UI.

IngestService turns PDFs into the gold layer. WorkspaceService records every canvas action as a versioned event. Both speak MCP, HTTP, and SSE.

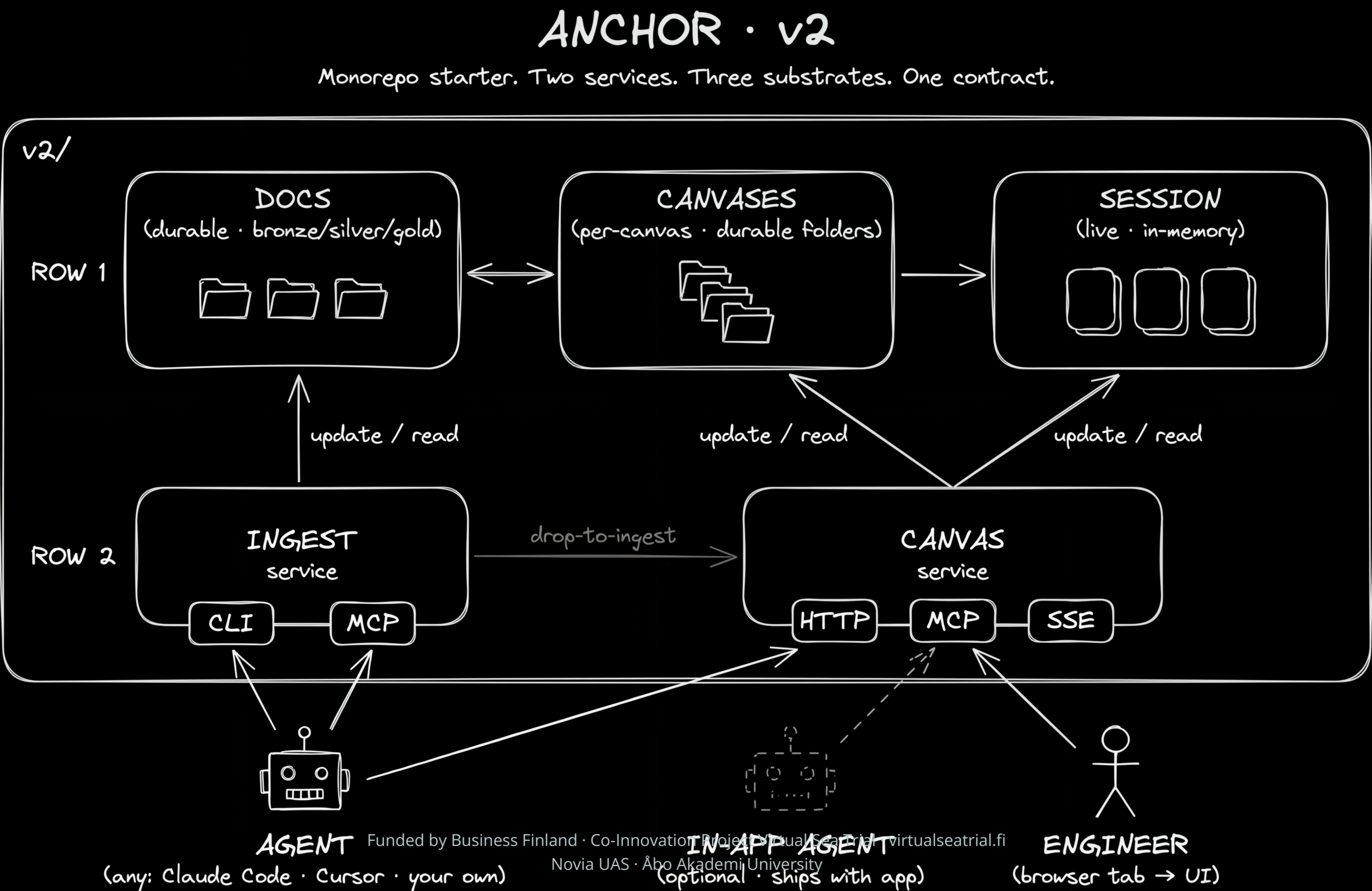
WHAT V0.1 SHIPS

- 17 MCP commands — 6 ingest, 11 canvas. Claude Code, Cursor, or your own agent drives the same data.
- Web UI in the same backend. Drop a PDF; ingest progress streams to every browser tab and agent.
- ~100 ms multi-client sync between agents and browsers.

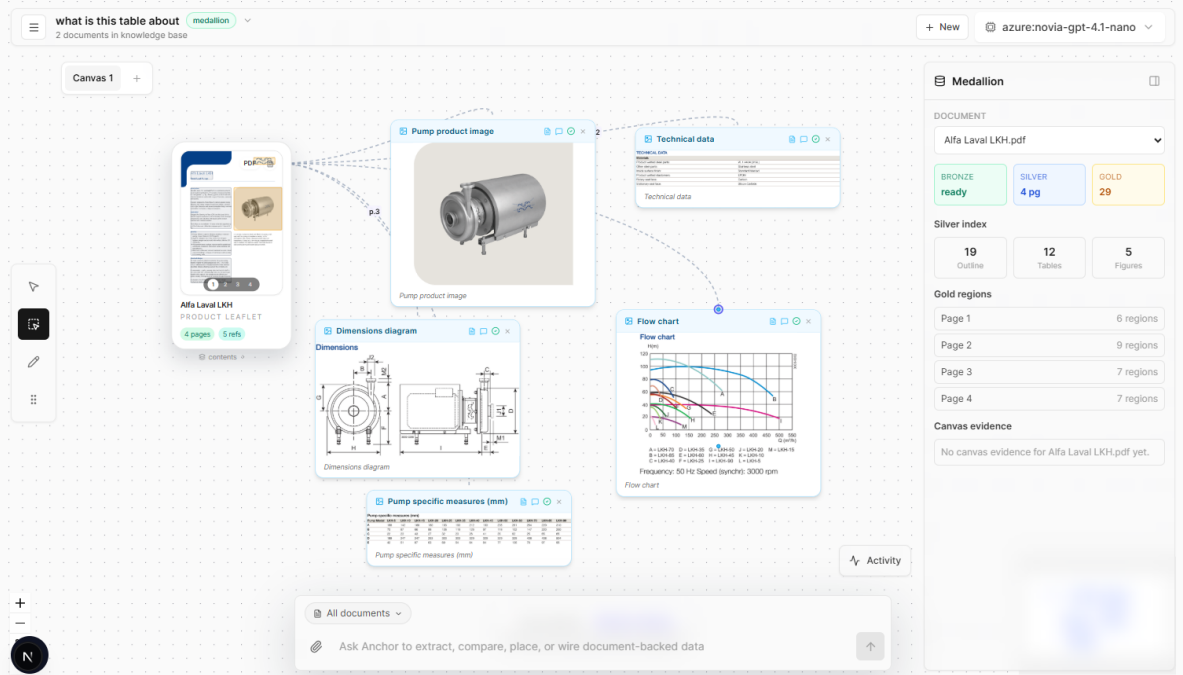
Novia University of Applied Sciences

Every spec your agent quotes
points back to the
page it came from.

Datasheet → canvas → simulation.
Provenance built in.



THE CANVAS



Question top. Cards carry source crops with one-click jump to the PDF region. Right rail: bronze → silver → gold ingestion of the active datasheet.

SELF-HOSTED

- Runs on a laptop, server, or air-gapped PC. No cloud account, no telemetry.
- Each canvas is a folder. Copy, share, version-control.
- Open source.

READ MORE



github.com/Novia-RDI-SeaFaring/
anchor-kb-ui-RAG

Industry presentation · Vaasa, 2026